

Element E

For our solution we incorporated the principles of STEM by using the engineering method, math, and technology to create the design and prototype. We had a great amount of thought and planning that went into the initial designs that we considered. Before we started to build the prototype we drew out the design that we settled on. Through a series of sketches we clearly communicated with each other on how we would build the prototype.

We then researched materials to use for the prototype looking at their different strengths and weaknesses. Through this process we compared the ease to use and manipulate the material, cost, and strength. After consulting with Mr Kiser about materials as well as thorough research we decided to use PVC as the material for the project. When we decided on that the drawings that we earlier made became very helpful as the parts we needed and needed to make became clear. We used math to figure the lengths that we needed for all the tubes that we used.

The Engineering Design Process came into the project helping us to decide on a design help make it and work all different parts to bring the project together. We made the initial prototype testing and making derivations along the way to better fit the cart and Mr. Kiser, the client. After we had a working prototype we started to test the strength of it finding that one of the joints in the middle cracked as it is a weak point.

After we had tested out prototype we had a day in class where we went with several engineers. Our group spoke with Peter Herman who is an engineer with salesforce. We presented the prototype with him and talked about the issue of strength while staying within the budget. We talked over multiple different materials including PVC, wood, and steel. In the end he suggested if we could have something that went inside of the existing handle we have that would strengthen it.

We also talked with Piper Cannon who is an engineering student at Vanderbilt. When we explained to Piper she agreed that we didn't need to rebuild the entire handle and just need something to go in the vertical part of the handle to give better strength and rigidity. We then found a couple steel pipes that fit inside the PVC. The problem that this brought up was the steel was too small to have a tight fit within the PVC so we brainstormed different ways to solve for that. The solution we settled on 3D printing small sleeves to go inside the PVC and tightly fit around the steel.

Both of the engineers we spoke with agreed that we had a good general design for the handle but it needed some small features and changes. The strength was an obvious issue that we worked with them on but Peter asked how we had made it adjustable as the criteria stated that. We said that for different carts of different widths you could change the horizontal pipes to larger or shorter lengths depending on what width the cart is. As for adjustability between people we explained that you could easily grab the vertical sections of the handle which has an ergonomic grip for people. Overall they agreed with the way we made our handle and the way that we got there.